STUDENT PERFORMANCE ANALYTICS SYSTEM

1. Introduction

The Student Performance Analytics System is a Python-based project designed to analyze student academic performance. It processes student records containing subject marks, department details, and attendance percentages.

2. Objectives

- Calculate total and average marks for each student.
- Assign grades based on average marks.
- Determine pass or fail status.
- Identify top-performing students.
- Calculate subject-wise and department-wise averages.
- Analyze grade distribution and attendance.

3. Technologies Used

- **Python:** Main programming language.
- **Pandas:** Used to read, organize, and analyze student data.
- **NumPy:** Used for numerical calculations.
- **CSV:** Used to store input and output records.

4. Dataset Description

The project uses a sample dataset containing 20 student records. The dataset includes student ID, name, department, Mathematics, Physics, Chemistry, English, Computer Science, and attendance percentage.

5. Implementation

The program loads student records from students.csv using Pandas. NumPy calculates total and average marks. Conditional statements assign grades and determine pass or fail status. Pandas functions analyze subject performance, department averages, grade distribution, and attendance.

The example grading criteria are:

- A: Average of 90 or above
- B: Average of 80 to below 90
- C: Average of 70 to below 80

- D: Average of 60 to below 70
- E: Average of 50 to below 60
- F: Average below 50

A student passes under the example rule when every subject mark is at least 40.

6. Results

The program generates a report containing student totals, averages, grades, and results. It also identifies top-performing students, compares subject and department averages, counts grades, and flags attendance below 75%.

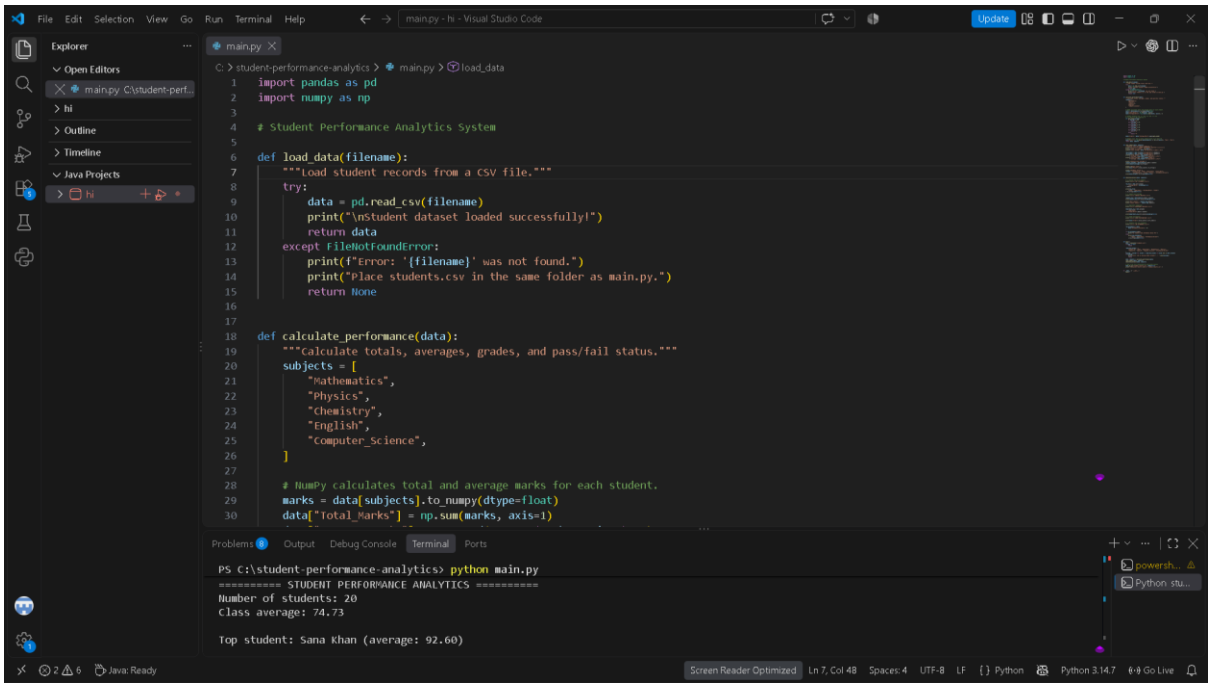
The analyzed records are saved to student_results.csv.

7. Conclusion

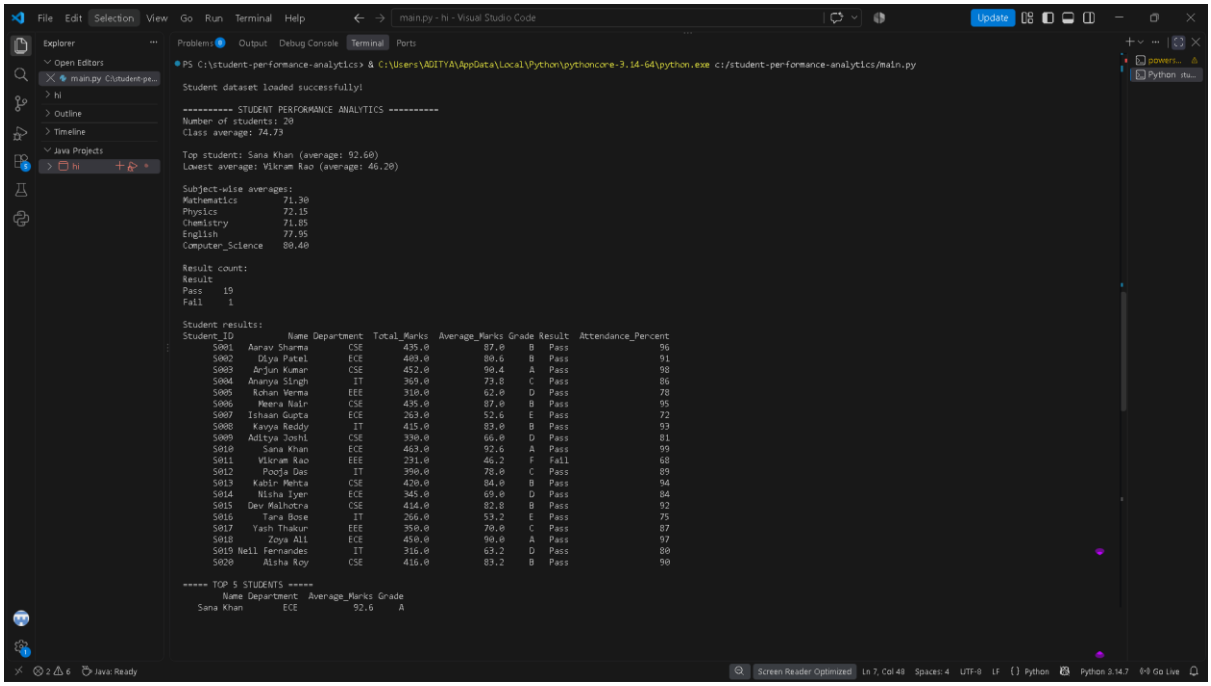
This project demonstrates the use of Python, Pandas, and NumPy to analyze student academic data. It provides a structured way to summarize marks, compare performance, and identify students who may need additional academic support.

8. Project Repository

GitHub Repository: https://github.com/aditya15128/student-performance-analytics



Python implementation of the Student Performance Analytics System.



```
PS C:\student-performance-analytics> & C:\Users\ADITYA\AppData\Local\Python\pythoncore-3.14-64\python.exe c:/student-performance-analytics/main.py

Student dataset loaded successfully!

---------- STUDENT PERFORMANCE ANALYTICS ----------
Number of students: 20
Class average: 74.73

Top student: Sana Khan (average: 92.60)
Lowest average: Vikram Rao (average: 46.20)

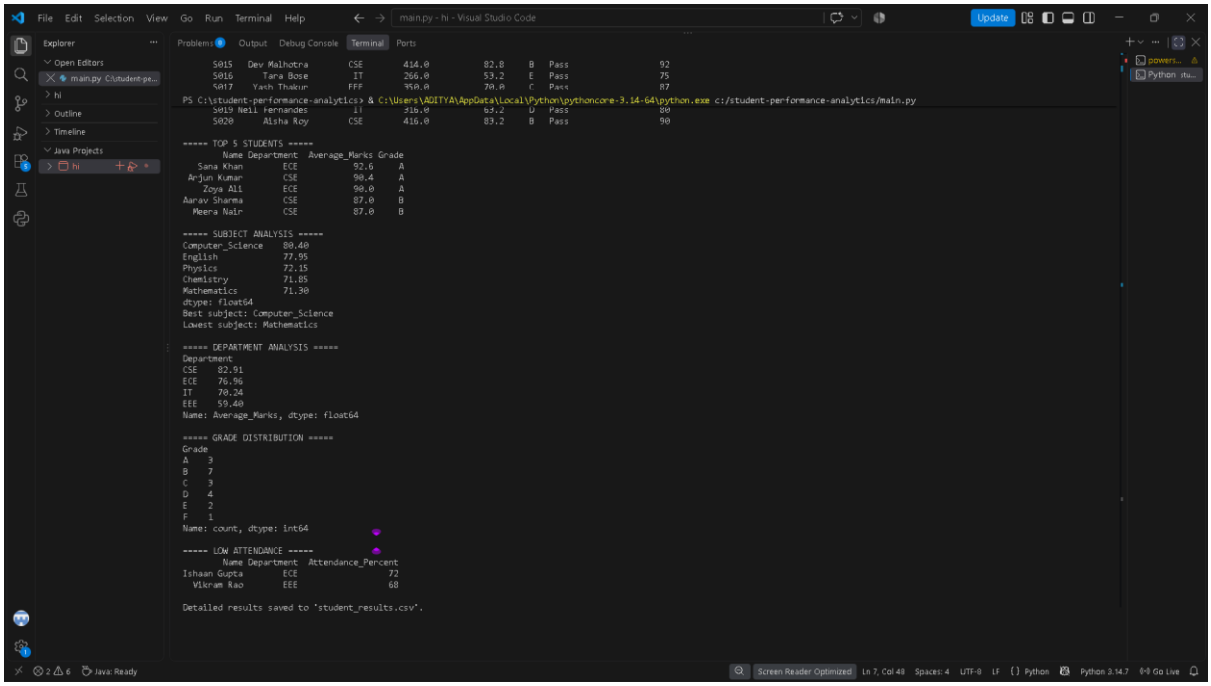
Subject-wise averages:
Mathematics         71.30
Physics             72.15
Chemistry           71.85
English             77.95
Computer_Science    80.40

Result count:
Result
Pass    19
Fail     1

Student results:
Student_ID          Name Department  Total_Marks  Average_Marks Grade Result  Attendance_Percent
      S001   Aarav Sharma        CSE        435.0           87.0     B   Pass                  96
      S002     Diya Patel        ECE        403.0           80.6     B   Pass                  91
      S003    Arjun Kumar        CSE        452.0           90.4     A   Pass                  98
      S004   Ananya Singh         IT        369.0           73.8     C   Pass                  86
      S005    Rohan Verma        EEE        310.0           62.0     D   Pass                  78
      S006     Meera Nair        CSE        435.0           87.0     B   Pass                  95
      S007   Ishaan Gupta        ECE        263.0           52.6     E   Pass                  72
      S008   Kavya Reddy         IT        415.0           83.0     B   Pass                  93
      S009   Aditya Joshi        CSE        330.0           66.0     D   Pass                  81
      S010     Sana Khan         ECE        463.0           92.6     A   Pass                  99
      S011    Vikram Rao         EEE        231.0           46.2     F   Fail                  68
      S012     Pooja Das          IT        390.0           78.0     C   Pass                  89
      S013   Kabir Mehta         CSE        420.0           84.0     B   Pass                  94
      S014    Nisha Iyer         ECE        345.0           69.0     D   Pass                  84
      S015   Dev Malhotra        CSE        414.0           82.8     B   Pass                  92
      S016     Tara Bose          IT        266.0           53.2     E   Pass                  75
      S017   Yash Thakur         EEE        350.0           70.0     C   Pass                  87
      S018      Zoya Ali         ECE        450.0           90.0     A   Pass                  97
      S019 Neil Fernandes         IT        316.0           63.2     D   Pass                  80
      S020     Alsha Roy          CSE        416.0           83.2     B   Pass                  90

===== TOP 5 STUDENTS =====
         Name Department  Average_Marks Grade
   Sana Khan        ECE           92.6     A
```

Student records and overall performance summary.



```
      S015   Dev Malhotra        CSE        414.0           82.8     B   Pass                  92
      S016     Tara Bose          IT        266.0           53.2     E   Pass                  75
      S017   Yash Thakur         EEE        350.0           70.0     C   Pass                  87
PS C:\student-performance-analytics> & C:\Users\ADITYA\AppData\Local\Python\pythoncore-3.14-64\python.exe c:/student-performance-analytics/main.py
      S019 Neil Fernandes         IT        316.0           63.2     D   Pass                  80
      S020     Alsha Roy          CSE        416.0           83.2     B   Pass                  90

===== TOP 5 STUDENTS =====
         Name Department  Average_Marks Grade
   Sana Khan        ECE           92.6     A
  Arjun Kumar       CSE           90.4     A
    Zoya Ali        ECE           90.0     A
Aarav Sharma        CSE           87.0     B
  Meera Nair        CSE           87.0     B

===== SUBJECT ANALYSIS =====
Computer_Science    80.40
English             77.95
Physics             72.15
Chemistry           71.85
Mathematics         71.30
dtype: float64
Best subject: Computer_Science
Lowest subject: Mathematics

===== DEPARTMENT ANALYSIS =====
Department
CSE    82.91
ECE    76.96
IT     70.24
EEE    59.40
Name: Average_Marks, dtype: float64

===== GRADE DISTRIBUTION =====
Grade
A    3
B    7
C    3
D    4
E    2
F    1
Name: count, dtype: int64

===== LOW ATTENDANCE =====
         Name Department  Attendance_Percent
Ishaan Gupta        ECE                  72
  Vikram Rao        EEE                  68

Detailed results saved to 'student_results.csv'.
```

Advanced analysis of student performance, grades, departments, and attendance.